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Double Standards

EDO, SDRAM, RDRAM and now DDR-SDRAM...the memory race is kicking up a lot of dust and DDR-SDRAM promises to be the new road warrior

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It was in 1997 that memory made the graceful transition from EDO to SDRAM. While this might seem like a small deal today, there were several issues that existed at that time. The interface was not backward compatible with existing motherboards and the performance it offered was only a shade better than what EDO then offered. Probably the single most important reason why the transition to the new memory type was relatively smooth was the fact that Intel had put its considerable weight behind it right from the beginning.

Today, about four years later, SDRAM reigns supreme in the memory market and has become the *de facto* standard. It has populated systems since the day the mighty 440LX/BX series motherboards were brought in by Intel to accompany the introduction of the Pentium II and the AGP standard to the latest chipset series. Other chipset and processor manufacturers have had a hard time just keeping up with this standard, and have fallen far short of

being able to dictate to or convince others about adopting a different standard.

Things changed dramatically with Intel's decision to adopt the radically new RDRAM solution. The release of the i820 chipset, the first to support the new memory standard, was a big step for Intel, aimed at pushing adoption of the new standard. However, the i820 chipset suffered major problems, which included several delays, and this coupled with high prices for the new memory chips meant that people weren't convinced that it was a viable solution.

Enter DDR-SDRAM. Double Data Rate (DDR) SDRAM was defined by JEDEC in 1997, and it was designed to be a natural migration from PC100 and PC133 SDRAMs to higher data rates.

Since its inception in early 1997, DDR-SDRAM has clearly come a long way. It has now garnered enough muscle and backing to eliminate SDRAM and RDRAM as competitors. Although recently RDRAM did try to dethrone it, it was a vain effort. RDRAM never really got

